Nanosyntax: state of the art and recent developments♣

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1. Introduction

One of the goals of current linguistic theory is to reduce grammatical variation to the properties of lexical items. This desideratum is known in the literature as the Borer-Chomsky conjecture (Borer 1984, Baker 2008: 353). The main idea behind the conjecture is that the operations of syntax are invariant across languages, variation arising solely from the interaction of those principles with language-specific lexical entries. This locates variation to a domain where the evidence for variation is indisputable -- the lexicon.

Nanosyntax shares this desideratum with current syntactic theories. However, the implementation of this idea in Nanosyntax differs from standard theories in one crucial respect. In Nanosyntax, the terminals of syntax are not language-specific feature bundles (i.e., pre-established assemblies of features), but individual features. Those features are furthermore hypothesised to be universal. There is thus no language-particular lexicon preceding the syntactic derivation. As a result, variation is restricted to the postsyntactic lexicon.

How can postsyntactic lexical entries trigger variation in the syntactic computation which precedes it? Starke (2014b; 2018) suggests that lexicalisation applies after every application of merge, and if it fails, the structure is handed back to syntax for last resort rescue operations — movements or complex left branch formation. In this model, the success of lexicalisation is thus dependent on the specific postsyntactic lexical items, and so are the last-

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resort movements. This resolves the timing tension of how postsyntactic entries influence what happens in syntax.

This mechanism proposed by Starke (2014b, 2018) is referred to as the *Lexicalisation Algorithm*. The Lexicalisation Algorithm regulates the process of externalization of syntactic structures. It takes the universal syntactic hierarchy or *functional sequence (fseq)* as an input and produces language-particular outputs due to its interaction with two other components of the grammar: the postsyntactic (language-particular) lexicon on the one hand and the (language-invariant) syntactic principles on the other hand. In effect, the algorithm steers the derivation in such a way that language-particular structures are formed as a function of language-particular (post-syntactic) lexicons. The prime source of variation has been identified as the size of a lexical entry, which refers to the number of features that can be lexicalised by a particular lexical item.

Work since 2018 has been marked by two important evolutions. The first of these is the realization that lexicalisation is not only affected by the size of lexical entries, but also by the structural arrangement of the features it contains. Two lexical entries of the same size may lead to different patterns of lexicalisation by organising the same features into different structures. The second innovation is more fundamental, in that it involves a change to the formulation of the Lexicalisation Algorithm itself. It is therefore natural for this volume to take the Lexicalisation Algorithm as its primary focus, and to explore its empirical effects across a wide range of phenomena. By assembling many different case studies under one roof, the volume showcases the wide applicability and the explanatory power of the algorithm.

In this chapter, we discuss the basics of phrasal lexicalisation in section 2, and relate it to Cinque's (2005) account of crosslinguistic differences in ordering. Section 3 explains how root size explains allomorph distribution. Section 4 introduces Starke's (2018) Lexicalisation Algorithm, and illustrates its operation using a concrete example. Section 5 discusses the shape of lexical items, in preparation of section Section 6, which discusses the recent update of the algorithm involving pied-piping and subextraction.

2. The basics of phrasal lexicalisation

Since its beginnings, one of the tenets of Nanosyntax has been to provide a theory of the syntax-phonology interface that is direct in the sense that the output of syntax is directly translated onto the phonological representation, without any intermediate morphological structure. A crucial ingredient that allowed Nanosyntax to progress towards this goal was the concept of phrasal lexicalisation, explored by Starke since the early 2000s (Starke 2002; see also McCawley 1968, Weerman and Evers-Vermeul 2002, Neeleman and Szendroi 2007). This notion refers to a scenario where phonological interpretation is assigned to a phrasal node containing one or more terminals.

While exploring this notion, it is important to keep in mind that the expressive power of phrasal lexicalisation is inextricably linked to the constituent structure provided by syntax. We therefore inevitably introduce the notion of phrasal lexicalisation along with the specific Nanosyntactic assumptions about syntactic structures. These draw upon the work in the cartographic tradition (Cinque & Rizzi 2010a), which assumes the existence of highly structured syntactic hierarchies (referred to as *fseq*, the functional sequence, in the Nanosyntactic literature). See for instance the decomposition of NP (Abney 1987), CP (Rizzi 1997, 2004), IP (Cinque 1999), PP (Koopman 2000, Svenonius 2010), NegP (Zanuttini 1991, 1997; Poletto 2008), TopicP (Benincà and Poletto 2004; Frascarelli 2007; Frascarelli and Hinterhölzl 2007), as well as other domains.

A final important inspiration of the Nanosyntax approach is the theory of possible orders proposed in Cinque (2005). Let us illustrate all three components (the theory of phrasal lexicalisation, the functional sequence, and the theory of ordering) by the example of Greenberg's Universal 39, given in (1).

(1) Greenberg's Universal 39 (Greenberg 1961: 95)

Where morphemes of both number and case are present and both follow or both precede the noun base, the expression of number almost always comes between the noun base and the expression of case.

This generalisation can be illustrated by languages such as Turkish in (2) or Tagalog in (3), where the number marker (in bold) intervenes between the root and the case, no matter whether we are looking at suffixes (as in (2)) or prefixes (as in (3)).

(2) Turkish

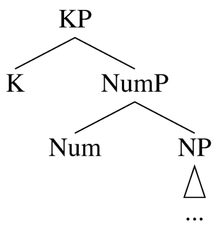
a.	el	-de		b.	el	**-ler**	-de
	hand	-LOC			hand	-PL	-LOC
	'in hand'				'in hands'		

(3) Tagalog

a.	sa	=bata		b.	sa	=**mga**	=bata
	DAT	=child			DAT	=PL	=child
	'to the child'				'to the children'		

The fact that languages tend to place number markers in between the noun and the case marker can be captured if we assume a crosslinguistically invariant hierarchy of functional projections (the functional sequence, *fseq*), where number is lower than case, and therefore, by default, closer to the noun, see (4).

(4)



Assuming a universal hierarchy like this disallows the generation of orders such as Num>K>N (with case in between number and the noun), because such an order can only arise from a hierarchy where Num is merged higher than K, violating the universal hierarchy of merge order. This ingredient of the cartographic approach is fully embraced in Nanosyntax.

Consider now the fact that Greenberg (1961) states the generalization in (1) as a tendency, which means that there are counterexamples. As a case in point, consider the Classical Armenian example in (5).

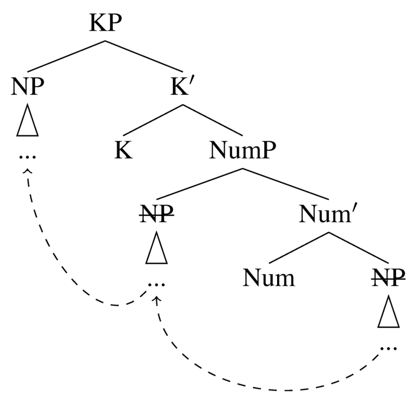
(5)	a.	azg	-aw	b.	azg	-aw	-k’
		nation	-INS		nation	-INS	-PL
		‘with a nation’			‘with nations’		

Relevantly, as reported in a typological study by Kloudová (2020), examples like (5) only appear in a position after the noun, and never in a position before the noun (i.e., the order Num>K>N is not found). This observation falls in line with the approach pioneered by Cinque (2005) for the ordering of nouns, adjectives demonstratives and numerals (Greenberg’s

Universal 20). In this domain, we also observe one order before the noun, and various different orders after the noun.

To account for this, Cinque (2005, 2009) adopts Kayne's proposal that the syntactic hierarchy is always mapped on a left-to-right linear order based on c-command. This produces the invariant order before the noun. For case and number morphology, the underlying structure in (4) thus maps onto the order K>Num>N. Any other order is derived by a leftward movement of a phrasal constituent containing the noun. In Cinque's approach, Classical Armenian thus has the structure as shown in (6).[1]

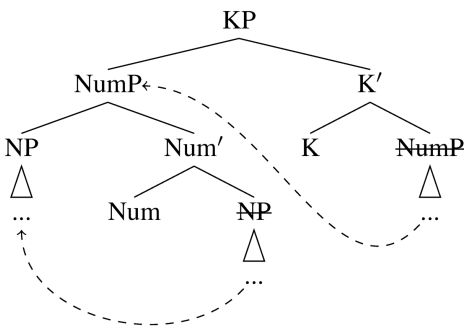
(6)



In (6), instead of staying in the base position, the noun contained in the NP moves cyclically across the projections of number and case without inverting their order, yielding the surface order N>K>Num. The first movement that brings the noun across Num will be referred to as complement movement because the whole complement moves across Num. The second movement will be referred to as a 'spec-to-spec' movement because only the complex left branch inside the complement of K moves across it. As we shall shortly see, these two movement types (complement movement and spec-to-spec movement) form the backbone of the Lexicalisation Algorithm proposed by Starke (2018).

We must, however, mention at this point that while Nanosyntax adopts the geometric properties of the trees proposed in Cinque's approach, there are also some differences that become important later. Specifically, in Cinque (2005), specifiers are dominated by a label related to their sister, their movement is triggered by attracting features, and they enter in agree(ment) relationships. In Nanosyntax, phrases that sit in the same geometrical position lack these properties: their movement is triggered by the need to lexicalise the syntactic structure (‘lexicalisation-driven movement’, also referred to as ‘evacuation movement’), a type of movement which doesn't project a label or leave a trace, contrary to feature-driven movements (see also Section 8). Furthermore, their mother node is not labelled. For now, this is merely a notational difference, but the unlabelled character of the nodes created by movement will become relevant in Section 6.

Consider now the Turkish order N>Num>K. This order would, under Cinque’s (2005) approach, be derived by the sequence of two complement movements, whereby we first move the NP across Num, and then again move the whole complement across K, as in (7).



(7)

In sum, to derive the attested orders, Cinque assumes that for each contentful head that is merged into the tree, there are three different derivational options. First, it can be the case that the complement of the head stays in situ and there is no movement. Second, the specifier of the complement can move, as when the NP moves across K in Classical Armenian. Third, the whole complement of the head can move, as when the NumP moves across K in Turkish. Nanosyntax adopts this general view on the derivation, even though it changes (as we shall discuss) some of the properties of the trees, e.g., the trigger for the movement and the labelling conventions.

With the basic assumptions concerning the functional sequence and the movements clarified, let us turn to phrasal lexicalisation. The role phrasal lexicalisation plays in Nanosyntax can be illustrated by means of a comparison between Latin and Turkish in Table .1.

	TURKISH 'HOUSE'		LATIN 'GRANDFATHER'	
	SG	PL	SG	PL
NOM	ev	ev-ler	av-us	av-i
ACC	ev-i	ev-ler-i	av-um	av-os
GEN	ev-in	ev-ler-in	av-i	av-orum

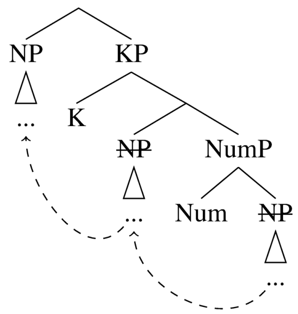
Table 1 Latin and Turkish number and Case morphemes

In Turkish, the morphology is agglutinative: the morpheme *-lAr* (with a harmonizing vowel) realizes PL, while *-i* realises accusative and *-In* genitive (see Türk and Caha 2021 for an analysis of the Turkish declension). In contrast, the Latin morphemes are fusional, i.e., they realise more than one grammatical feature. There is no way of segmenting the Latin morphemes into invariant plural or case markers.

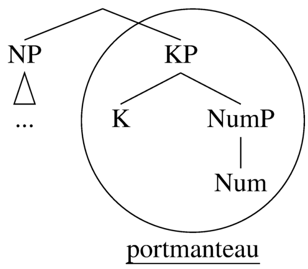
In theories where lexicalisation only targets terminal nodes, the Latin facts present a paradox. One branch of the paradox stems from Greenberg's Universal 39, which requires number and gender to be independent projections, universally ordered (e.g. by *fseq*). The other branch of the paradox stems from Latin: under terminal-node lexicalisation, Latin suggests that there are no two independent projections, but rather a single projection hosting the Latin morpheme combining case and number. Put differently, if there are two independent projections, the Latin morpheme is spread across several terminals, contradicting the terminal-based approach.

That paradox is resolved in Nanosyntax, without invoking any mechanism dedicated to it: the information spread across several terminals corresponds to a phrase, and phrasal lexicalisation will target that constituent – no paradox. Specifically, the idea is that the noun in Latin moves cyclically to the top of the tree, as shown in (8)(a). Note that since we now switch to a Nanosyntax account, we also change the labelling conventions, with unlabelled nodes dominating phrases which are similar to traditional "specifiers". After the movement, we obtain (8b), a structure where K and Num form a constituent (setting aside the issue of traces of the moved noun). This constituent is lexicalised by the portmanteau morpheme.

(8) a.



b.



Phrasal lexicalisation extends the notion of constituency to a new domain: lexicalisation. When multiple terminals jointly move to a particular position (e.g., to the left of the verb in V2 languages), it is assumed that a whole constituent containing all the relevant terminals moves as one unit, i.e., as a constituent. When multiple terminals are not pronounced due to ellipsis, it is assumed that the ellipsis targets the full phrase containing these terminals (e.g., VP ellipsis). The idea in (8)(b) extends this rationale to the pronunciation of multiple terminals: when multiple terminals are realised by a single morpheme, this is because the morpheme lexicalises a constituent containing these terminals.

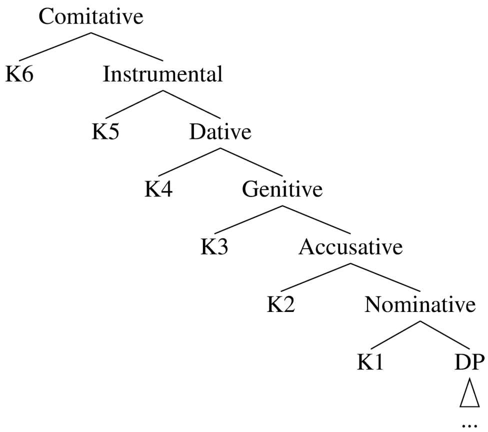
Summarising, the discussion above shows that phrasal lexicalisation in combination with the types of movements utilized in Cinque (2005) allows us to capture the differences between the number of terminals and the number of morphemes. It does so without altering the underlying syntactic structure (number and case remain separate terminals) and therefore allows one to maintain the idea of a universal functional sequence. It also does not require any postsyntactic operation (like Fusion, an operation used in Distributed Morphology (Harley and Noyer 1999)), thereby also avoiding the need for a dedicated morphological structure, distinct from syntax.

Nanosyntax generalizes this logic beyond obvious examples of fusional and portmanteau morphology, such as the Latin genitive plural *-orum* above. To see this, consider agglutinative case markers, such as the Turkish genitive *-In* or the Armenian instrumental *-aw*: they also lexicalise with a single morpheme what other languages lexicalise with more than one morpheme. In Caha's (2010) containment data (Table) for instance, the genitive, which is lexicalised as a single morpheme in Turkish (as we saw in Table 1.1) is lexicalised by two suffixes in Vlakh Romani (the accusative suffix followed by *kor*), and by a preposition together with an accusative pronoun in English. Similarly for the instrumental, which is lexicalised as a single morpheme in Classical Armenian (see (5a)), but by two morphemes in Budukh and German.

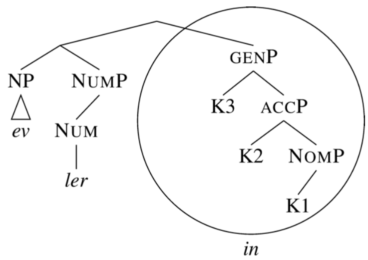
Case	**Language**	**Expression**		
		P	SUFF1	SUFF2
GEN	Vlakh Romani		ACC	*kor*
	English	*Of*	ACC	
DAT	Estonian		GEN	*le*
	Arabic	*Li*	GEN	
INS	Budukh		DAT	*Vn*
	German	*Mit*	DAT	
COM	Georgian		INS	*gan*
	Russian	*S*	INS	

Table 2 Case containment relations (from Caha 2010)

Based on this containment, syncretism patterns, and a number of other facts, Caha (2009, 2010) proposes a hierarchy of cases, such that every containment relation in Table is reflected by structural containment in the functional sequence: the comitative case is built by adding a feature (K6) to the instrumental, the instrumental is built by adding a feature (K5) to the dative, and so on, see (9).

(9)


Once this decomposition is adopted, agglutinative languages with no overt case containment, like Turkish, come out similar to Latin, in that they possess a case portmanteau that lexicalises all the relevant features. This is shown in (10), where it is assumed that NP first moves across Num, and then the entire NumP cyclically moves across the case features.

(10)


For more discussion, see Caha (2009), where it is argued that phrasal lexicalisation (based on a particular matching condition and incorporating the Elsewhere Condition) derives the fact that syncretism between case markers targets adjacent projections in the *fseq*, see (11). Such statements are referred to as the *ABA generalisation (cf. Baerman et al. 2005, Bobaljik 2012, Smith et al. 2019).

(11) In the sequence *nominative – accusative – genitive – dative– instrumental – comitative,* syncretism targets only adjacent cases (Caha 2009: 10).

The study of syncretisms (as in (11)) and containment (as in Table 1.2) led to a rich body of work investigating the structure of the *fseq* in the nominal domain (Caha 2009; 2022; this volume; Taraldsen 2010; Starke 2017; Bergsma 2019; Kloudová 2020; Türk and Caha 2021; Janků 2022), directional expressions (Pantcheva 2011), negation (De Clercq 2013; 2020; Baunaz and Lander 2023), demonstratives (Lander 2016; Lander and Haegeman 2018), pronominals (Vanden Wyngaerd 2018), complementizers (Baunaz 2018; Baunaz and Lander 2018a; 2018c), cross-categorial syncretisms (Wiland 2019), the inflectional and derivational morphology of verbs (Taraldsen Medová and Wiland 2018; Baunaz and Lander 2019; Starke and Cortiula 2021; Cortiula 2023), degrees (Caha, De Clercq, Vanden Wyngaerd 2019, Vyshnevska 2022, Wiland 2023) or indefinites (Dekier 2021). The investigation of the *fseq* by means of the study of syncretisms and morphological and semantic containment is ongoing, and in this volume, the chapters by Dikmen and Demirok, and by Baunaz and Lander add to this investigation, too. We discuss the issue of syncretism and *ABA in more detail in sections 5 and 6.

3. Root size as an allomorphy trigger

This section introduces the idea that not only functional morphemes, but also roots may differ in how many grammatical features they realise, and in doing so, give rise to allomorphic variation in suffixation patterns. For example, the roots *men* or *sheep* are lexically associated to plural, while the roots *boy* or *dog* are not. This is why the latter but not the former need a plural suffix. This section provides analogous examples from the comparison of Czech, English and Armenian degree morphology.

To introduce the basic idea, consider first Czech in (12). The examples show an increasingly complex morphology as we go from the positive (*intelligent-n-í* 'intelligent') to the comparative (*intelligent-n-**ějš**-í* 'more intelligent') to the superlative (***nej**-inteligent-n-ějš-í* 'most intelligent'). The extra morpheme is always in bold in (12).

(12) a. Emm-a je inteligent-n-í. POSITIVE

Emma-NOM be.3SG.PRS smart-N-FEM.SG

'Emma is intelligent.'

b. Emm-a je inteligent-n-**ějš**-í než Pavel. CMPR

Emma-NOM be.3SG.PRS smart-N-CMPR-FEM.SG than Pavel

'Emma is more intelligent than Pavel.'

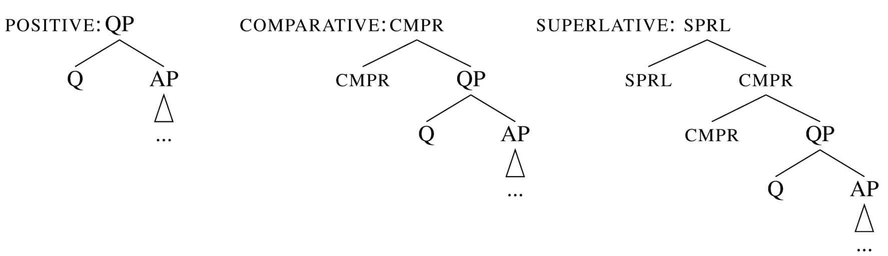
c. Emm-a je **nej**-inteligent-n-ějš-í ze všech. SPRL

Emma-NOM be.3SG.PRS SPRL-smart-N-CMPR-FEM.SG from all

'Emma is the most intelligent of all.'

The increasing complexity of the morphology can be captured under the proposal (due to Bobaljik 2012) that the superlative degree contains the comparative, which is in turn based on the positive, see (13); recall also the discussion of containment in Section 2.

(13)



The positive degree as depicted in (13) is assumed to be bi-componential (enriching slightly Bobaljik's original proposal). On top of the lexical AP, the positive contains a QP, a functional category borrowed from Bresnan's (1973) and Corver's (1997) work that encodes the gradability/scalarity of the adjectives under discussion (see De Clercq et al. 2023 for discussion). The immediate motivation for it is to capture the fact that the adjective *inteligent-n-í* 'intelligent' is bi-componential in Czech (disregarding the agreement), which shows the need to assume at least two positions of exponence in the positive (see Vanden Wyngaerd et al., to appear, for argumentation).

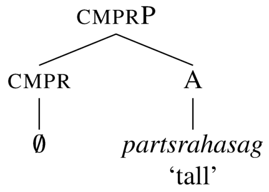
Czech contrasts with Armenian (Bobaljik 2012: 88, 91), where the form of the adjective is the same in all degrees, see (14): the adjective for 'tall' is *partsrahasag* regardless of the degree.

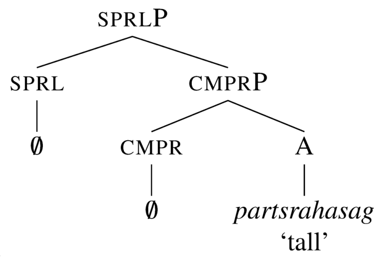
(14) a. Artak-ə **partsrahasag** e. POSITIVE
Artak-DEF tall be.3SG.PRS
'Artak is tall.'

b. Artak-ə Bagrat-e-n **partsrahasag** e. CMPR
Artak-DEF Bagrat-ABL-DEF tall be.3SG.PRS
'Artak is taller than Bagrat.'

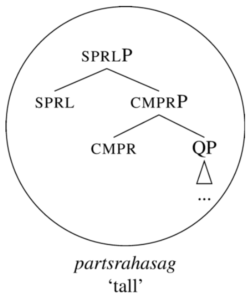
c. Artak-ə amen-e-n **partsrahasag** e. SPRL
Artak-DEF all-ABL-DEF tall be.3SG.PRS
'Artak is the tallest.' (lit. 'Artak is taller than all.')

Assuming that the underlying functional sequence is invariant, a terminal-only insertion would force one into assuming multiple zeroes, as in (15).

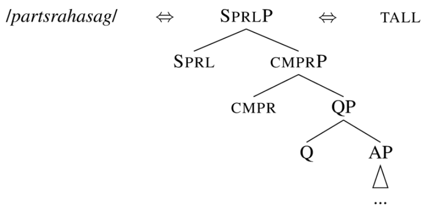
(15) a. 

b. 

However, if one adopts phrasal lexicalisation, then one could say that Armenian roots like *partsrahasag* 'tall' can lexicalise all features relevant to the adjectival root (QP), as well as the comparative and the superlative, as in (16).

(16) 

At this point, we need to say something about how lexical items are stored in the lexicon. The general assumption is that lexical items are stored (memorised) links between representations belonging to different modules. For example, the depiction in (16) corresponds to the syntactic representation of the superlative. In the lexical entry (17), the syntactic tree is linked to a phonological representation (between slanted brackets) and conceptual information (in small capitals). In essence, the entry says that if syntax builds a structure like the one in (17), this structure can be linked to the phonological representation */partsrahasag/* and the concept TALL.

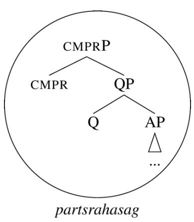
(17) 

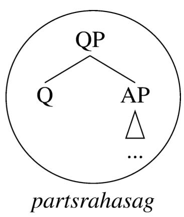
To determine whether a lexical entry matches a syntactic structure, Nanosyntax relies on identity. Specifically, if the constituent built by syntax is identical to a constituent stored in the lexicon, the entry of that constituent can be used to 'lexicalise' the syntactic representation; as a result, the relevant syntactic structure can be linked to a particular phonological and/or conceptual representation. This condition on matching is given in (18).

(18) *Condition on matching*

A lexically stored constituent L matches a syntactic phrase S iff S is identical to L.

The condition makes it possible for the lexical item in (17) to lexicalise either of the syntactic structures in (19), i.e. those of the comparative and the positive. This is so because any subconstituent of a lexically stored tree is itself stored in the lexicon, and therefore both the CMPRP constituent of (19)(a) and the QP constituent of (19)(b) find a matching constituent in the entry (17).

(19) a. 

b. 

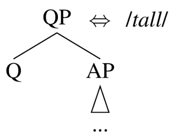
Phrasal lexicalisation, understood as identity matching, thus derives the syncretism between the three degrees of comparison that we observe in Armenian without any additional assumptions.

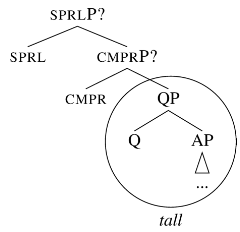
The matching condition (18) is an unpacked version of the matching condition used in Starke (2009): 'A lexically stored tree matches a syntactic node iff the lexically stored tree contains the syntactic node'. To show that, let us first expand it to: 'A lexically stored tree matches a syntactic node iff the lexically stored tree contains a constituent identical to the syntactic node'. This formulation further implicitly assumes that we look at the root constituent of the lexical tree, and because of that, the definition has to explicitly allow matching by constituents that it 'contains'. Removing that implicit stipulation allows a simplified formulation: 'A lexically stored tree matches a syntactic node iff the lexically stored tree is identical to the syntactic node', i.e. (18). The goal of unpacking the original formulation is to make the identity requirement on matching explicit.

The matching condition (18) -- just like the original formulation -- derives the Superset effect, meaning that a lexical entry can lexicalise any phrase that it contains. Because of this, the matching condition is sometimes referred to as the Superset Principle (also in the subsequent chapters). However, it is important to realise that the Superset Principle is not an independent principle in addition to the matching condition (they are two different names for the same thing).

Let us now turn to English, which, unlike Armenian, requires additional morphology in the comparative and superlative (*tall* – *tall-**er*** – *tall-**est***). If the lexical tree for *tall* is as in (20)(a) (omitting the conceptual information), we can understand why this is so: the lexical tree of *tall* is too small to lexicalize the degree features CMPR and SPRL, as (20)(b) shows. It can only realize the features of AP and QP. In effect, the root size of *tall* determines that the adjective will need help from other lexical items to lexicalize CMPR and SPRL.

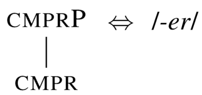
(20) a. lexical entry b. syntax

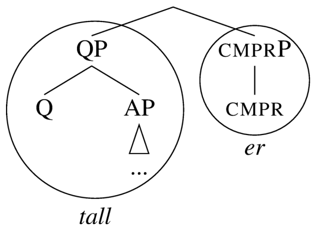




To capture the morphology of the forms *tall-er* and *tall-est*, we need two more assumptions. The first assumption concerns the morphosyntactic structure, namely, we need to assume that the QP in English moves across CMPR by complement movement, yielding the structure in (21)(b). In this structure, the CMPR feature is lexicalised as a part of the CMPRP, using the entry in (21)(a).

(21) a. lexical tree b. syntax

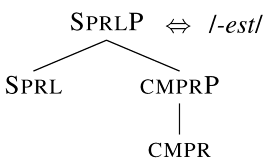


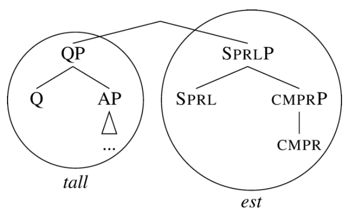


In the syntactic tree (21b), we assume the same types of lexicalisation-driven evacuation movements as introduced in Section 2, which leave no trace, and do not project a label (see the discussion in section 2).

In (22) we show an analogous analysis for the superlative.

(22) a. lexical tree b. syntax



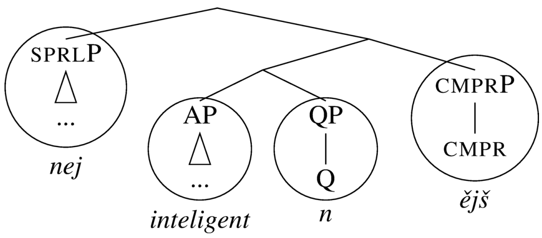


Note that as a consequence of the matching condition (18), the superlative *-est* with the lexical entry as in (22)(a) is a candidate for the lexicalisation of the comparative in (21)(b) because the CMPRP in (21b) is identical to a subtree of (22)(a). The reason why *-er* nevertheless appears in the comparative is that the lexical tree of *-er* in (21)(a) is a better match for the structure in (21)(b) as it has fewer superfluous features compared to (22)(a). In sum, when there are multiple candidates, they compete, and the winner is the lexical entry that is the best match with the syntactic structure. This competition principle is an instance of the Elsewhere Condition, going back to Kiparsky (1973):[2]

(23) In case two rules, R1 and R2, can apply in an environment E, R1 takes precedence over R2 if it applies in a proper subset of environments compared to R2.

Let us now come back to the pattern of the Czech adjective *inteligent-n-í* 'intelligent,' with a comparative form *inteligent-n-**ějš**-í*, and a superlative ***nej**-inteligent-n-ějš-í*. Czech differs from English on two counts. First, unlike in English, the superlative marker does not replace ('override') the comparative, but rather attaches on top of the comparative. Intuitively, this is because the Czech morpheme *nej-* only lexicalises the feature SPRL, i.e. it is smaller than the English *-est*. This will require *-ejš-* to realize the CMPR feature in the superlative.

The second difference is that the positive in Czech is bimorphemic, consisting of the root and an adjectiviser *-n.* We may thus informally represent the situation in Czech as in (24).

(24)


Here the root only lexicalises the AP, which moves across Q by complement movement, and the adjectiviser lexicalises the QP. The whole QP then moves again across CMPR, and the CMPRP is lexicalised by the comparative *-ějš*. The lexical entry of *nej-* is unable to lexicalise the CMPR head realised by *-ějš*, and so we see the superlative prefix attaching to the comparative, which is unlike what happens with the English *-est*.

Summarising, this section so far showed that varying root size across languages (i.e., whether the root lexicalises AP, QP, or SPRLP) correlates with whether we find affixal material on the adjective in the positive, comparative and/or superlative. This can be captured if the root always lexicalises whichever features its lexical entry allows it to, and the remaining features are lexicalised by affixes or independent words.

It is important to observe that this type of variation is not only found across languages, but also within languages. For example, the English positive-comparative pair *bad* ~ *worse* has a suppletive root and a zero allomorph of the comparative. In Nanosyntax, the most natural analysis is one where there is no zero, rather *worse* has the same comparative structure as the Armenian *partsrahasag* 'tall,' i.e., containing no movement, recall (19)(a). This differs from *tall-er*, which requires complement movement, recall (21)(b). If that is so, we end up in a situation where different adjectives require different structures within the same language, i.e., English has both (19)(a) (no movement) and (21b) (complement movement), but for different roots.

Similarly, there are adjectives in Czech that do not require an adjectivizer, and their positive is monomorphemic, e.g., *chytr-ý* 'smart' with the comparative *chytř-ejš-í* 'smarter'.

Their structure must be therefore like the one of *tall* ~ *tall-er*, where the root lexicalizes the whole QP. Once again, this means that there is no movement of AP across Q with these adjectives, and we are once again led to conclude that different roots require different structures.

A compact way of representing and comparing the sizes of different lexical items is in the form of a lexicalisation table. The discussion of the degrees of comparison in Armenian, Czech, and English in this section, for example, can be summarised in a single overview in the form of a lexicalisation (Table). The top row of the table (following the language and the degree), represents the ingredients/features of the functional sequence, going from bottom to top as one reads from left to right. Actual morphemes that lexicalise these features are shown in the relevant cells of the table. Root size can be directly read off the table: the more cells a morpheme occupies, the larger its size. Black cells indicate features that are absent (e.g. the features CMPR and SPRL are not present in the positive).

<table>
<tr><th>language</th><th>degree</th><th>AP</th><th>Q</th><th>CMPR</th><th>SPRL</th></tr>
<tr><td>Czech</td><td>POS</td><td>intelligent</td><td>n</td><td colspan="2"></td></tr>
<tr><td>Czech</td><td>CMPR</td><td>intelligent</td><td>n</td><td>ějš</td><td></td></tr>
<tr><td>Czech</td><td>CMPR</td><td colspan="2">chytr</td><td>ějš</td><td></td></tr>
<tr><td>Czech</td><td>SPRL</td><td>intelligent</td><td>n</td><td>ějš</td><td>nej</td></tr>
<tr><td>English</td><td>POS</td><td colspan="2">smart</td><td colspan="2"></td></tr>
<tr><td>English</td><td>CMPR</td><td colspan="2">smart</td><td>er</td><td></td></tr>
<tr><td>English</td><td>CMPR</td><td colspan="3">worse</td><td></td></tr>
<tr><td>English</td><td>SPRL</td><td colspan="2">smart</td><td colspan="2">est</td></tr>
<tr><td>Armenian</td><td>POS</td><td colspan="2">partsrahasag</td><td colspan="2"></td></tr>
<tr><td>Armenian</td><td>CMPR</td><td colspan="3">partsrahasag</td><td></td></tr>
<tr><td>Armenian</td><td>SPRL</td><td colspan="4">partsrahasag</td></tr>
</table>

Table 3: Lexicalisation table of the degrees of comparison in Czech, English, and Armenian

The lexicalisation table also shows how different root sizes yield different patterns of affixation. The concept of root (and affix) size therefore neatly captures the differences between classes of adjectives within a language, as well as the differences between different root classes across languages. In this volume, such differences play a crucial role in the contributions by Bergsma, Don, Merkuur and Smith, by Caha and Taraldsen Medová, by Natvig, Putnam and Wilson, by Taraldsen, and by Vyshnevska.

When there are multiple ways of lexicalising the same set of features (e.g., *worse* vs. **badd-er*), the system of lexicalisation is set up in such a way that a preference for one type of lexicalisation against another type follows from the system. More concretely, Starke (2018) proposes that the movements which give rise to different structures are not triggered by syntactic features, but by the need to create a structure that can be lexicalized, i.e., externalised at the interface. The theory of how movement (and structure building in general) interacts with lexicalisation is the main content of the Lexicalisation Algorithm, which is in turn the main focus of this book. The Lexicalisation Algorithm interacts with lexical items in a way that different lexical items give rise to different structures within the same language. We provide more detail in the next section.

4. The Lexicalisation Algorithm

4.1. Five steps

The first important assumption is that features are merged one by one, and that whenever a new feature is merged, lexicalisation must first succeed before the next feature can be merged. To this end, the structure created by Merge is matched against the existing lexical entries of the particular language. When a matching item is found for the newly formed structure, the derivation may continue by Merge F. However, when no matching item is found, syntax tries to rescue the phrase marker by performing left-branch movement (akin to spec-to-spec movement in traditional theories) and lexicalising the result. If that fails, complement movement is attempted.

These steps are given in (25), and the first three of them (25)(a-b-c) quite straightforwardly correspond to Cinque's (2005) parameters of variation (no movement in (25)(a), spec movement in (25)(b), complement movement in (25)(c)) -- though in a different order of preference.

(25) a. Merge F and lexicalize.

b. If fail, try a spec-to-spec movement and lexicalize.

c. If fail, try a movement of the complement of the newly inserted feature and lexicalize.

d. If fail, go back to the previous cycle, and try the next option for that cycle.

e. If fail, spawn a new derivation providing feature X and merge that with the current derivation, projecting feature X to the top node.

(adapted from Starke 2018)

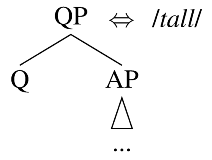
The ordering of the first option of the algorithm with respect to the others sets preferences for particular structures: lexicalisation without movement, whenever possible, is preferrable to movements: this will ensure that *worse* is preferred to *badd-er*. In what follows, we discuss the movement options of the algorithm (25)(b-c) (section 4.2), backtracking (section 4.3), and complex specifiers (section 4.4).

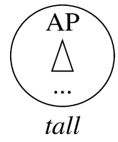
4.2. Movement

Turning first to the first three options of the algorithm (25)(a-b-c), let us see how English *tall* differs from the Czech *inteligent-n-í* 'intelligent' in derivational terms. The syntactic derivation starts by assembling AP, see (26)(a) (we assume that the AP is internally complex, as in Vanden

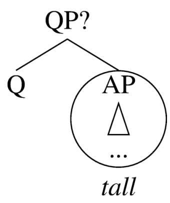
Wyngaerd et al. 2020). After this, the lexicon is checked, and since *tall* (repeated for convenience in (26)(b)) contains the AP, it is inserted, see (26)(c).[3]

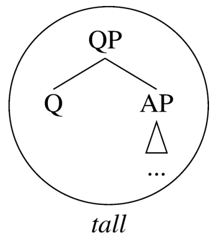
(26) a. 

b. 

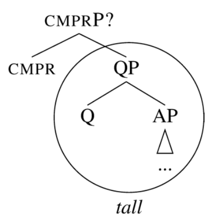
c. 

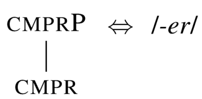
Since matching succeeded without any movement, the gradability feature Q can be merged, yielding (27)(a). (26)(b) is a perfect match and hence can lexicalize the structure, shown in (27)(b). This is the correct lexicalisation of the positive degree.

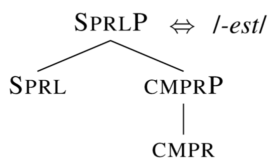
(27) a. 

b. 

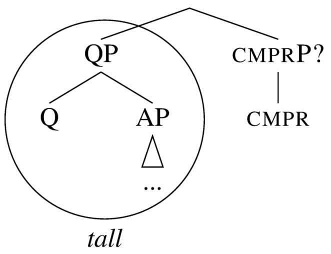
If deriving the positive degree was intended, the derivation would terminate (or merge the adjective with a noun). If the comparative is intended, syntax merges CMPR in the next step, see (28a). The lexicon is checked, but there are no lexical items matching this structure: the lexical tree for *tall* in (26)(b) does not contain the syntactic tree (28)(a). The lexical entries for *-er* and *-est* (repeated in (28)(b-c)) have CMPR, but neither of them matches the structure in (28a).

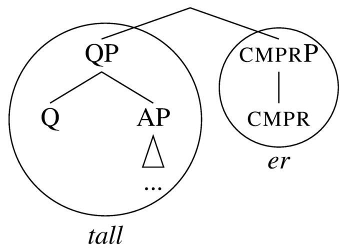
(28) a. 

b. 

c. 

Since no match is found, the syntactic tree (28)(a) cannot be externalised. This leads to rescue movements, following the Lexicalisation Algorithm in (25). The first movement to be tried is moving the spec of the complement, recall (25)(b). However, the complement of CMPR has no spec in (28)(a), so syntax proceeds directly to complement movement (moving QP out of CMPRP), yielding (29)(a). In (29)(a), CMPRP can be lexicalized by the lexical item (28)(b), see (29)(b).

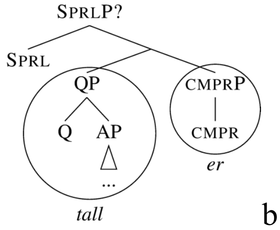
(29) a.


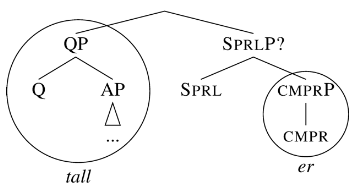
b.


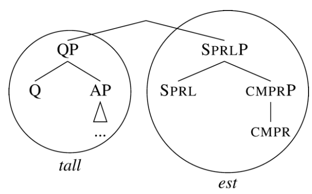
Summarising, movement is necessary if the constituent created by Merge does not find an immediate match in the lexicon. This movement is not a feature-driven, but it applies for the sake of lexicalisation: it is a lexicalisation-driven movement.

Because of this, the algorithm easily handles the fact that different roots interact differently with the derivation: for instance, if the root *worse* (capable of pronouncing (28)(a)) was compatible with the intended conceptual structure, (28)(a) would have been successfully lexicalised, and no movement would have been triggered.

Going back to our derivation of *tall*, consider now how the superlative is derived. First, syntax merges the feature SPRL to (29)(b), yielding (30)(a).

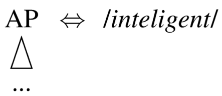
(30) a.


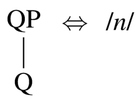
b.


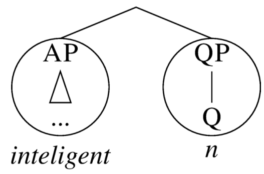
c.


Since lexicalisation is rigidly cyclic, the lexicon is checked again, but there is no matching lexical item for the structure in (30)(a). Therefore, syntax tries moving the specifier (recall (25)(b)), yielding the tree (30)(b). After QP moves out, the SPRLP on the right branch can be lexicalized by the lexical item in (28)(c), overriding the previous lexicalisation, i.e. *-er*, and leading to the structure in (30)(c).

Let us now turn to Czech. In this language, some adjectives (e.g., *chytr-ý* 'smart') are rather similar to *tall* and do not require any extra morpheme in the positive. The morphology of such adjectives follows if they lexicalise QP, like the English *tall.* However, recall that the adjective *inteligent-n-í* 'intelligent' requires a special adjectiviser in the positive, namely *-n*. We can model this by attributing the size AP to the adjectival root *inteligent-* and letting *-n* lexicalise QP, as shown in the lexical entries in (31)(a-b).

(31) a. 

b. 

c. 

Since (31)(a) cannot lexicalise the entire structure of the positive, AP has to move out of QP, yielding (31)(c), where the adjectiviser is inserted as the lexicalisation of the QP.

Summarising, we have now seen how the lexical items that a language has at its disposal 'decide' which of the three different movement scenarios in (25a-b-c) takes place. Given the appropriate lexical item, Merge can be followed straightaway by lexicalisation without movement as in (27)(b), but it can also be followed spec-to-spec movement (30)(b), or complement movement (31)(c). These are still the same movement types as proposed in Cinque (2005), but which of these applies is not dependent on 'strong features,' EPP, OCC and the like, but it depends on the lexical items available in each particular language – an explicit and detailed implementation of the Borer-Chomsky conjecture. Two additional steps of the algorithm remain to be discussed now: backtracking, (25d) and complex left branch formation (25e).

4.3. Backtracking

Backtracking corresponds to step (25d) in the Lexicalisation algorithm. This step is tried when the movement steps that we have discussed so far do not lead to successful lexicalisation. When the derivation reaches a dead end in this way, it returns to the previous cycle, and attempts the next option of the algorithm. To give a feel for the kind of phenomena that give rise to a treatment in terms of backtracking, we turn to a new empirical case, the nominal declension in Czech, as discussed in Janků (2022). There are a bit less than 20 different ls, depending on the noun class. We discuss two of them here, and only three different cases, as shown in Table 1.4.

	'bread'	'tongue'
NOM	chleb-a	jazyk
ACC	chleb-a	jazyk
GEN	chleb-a	jazyk-a

Table 4: Partial declension paradigms for two Czech noun classes

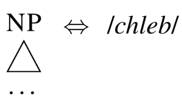
The nominative and the accusative show a contrast between the *chleba* 'bread' and *jazyk* 'tongue' class that is straightforwardly amenable to a treatment in terms of varying root size. The fact that nouns of the *jazyk* class have no suffix in the nominative and the accusative suggests that they have a big enough root to lexicalise the entire functional sequence up to and including the accusative. The nouns of the *chleba* 'bread' class are smaller, and require a suffix -*a* in the nominative and the accusative (assuming, as argued in Caha 2009 and explained section 2, that the different cases involve features like K1, K2, etc., which stand in a containment relation). Table 1.5 (somewhat simplified from Janků 2022) shows this in the form of a lexicalisation table.

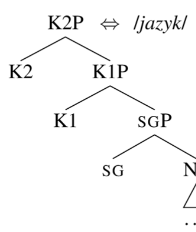
	NP	SG	K1	K2	K3
NOM	chleb	a			
ACC	chleb	a			
GEN	chleb	a			
NOM	jazyk				
ACC	jazyk				
GEN	jazyk	a			

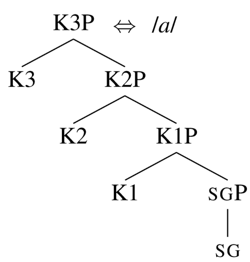
Table 5: Lexicalisation table for *chleba* 'bread' and *jazyk* 'tongue'

The surprise comes in the genitive, where the distinction between the two noun classes disappears, and both get the same suffix *-a.* This phenomenon of paradigm levelling is pervasive in Czech, where typically oblique cases show less noun class distinctions than structural cases. Informally put, it looks as if in the genitive, the noun *jazyk* shrinks to become the same size as that of the root *chleb*, requiring it to have the same suffix. This is shown schematically in Table 1.5 (see in particular the lines of GEN). The important point is that this shrinking of the root below its full lexicalisation potential is exactly what happens in a backtracking derivation.

Let us see in some more detail how this works. In (32) we show the lexical entries for the roots *chleb-, jazyk-*, and the suffix *-a*, respectively.

(32) a. 

b. 

c. 

The derivation for *chleba* is straightforward and does not introduce anything new beyond what we discussed for the derivation of adjectives above. The same is true for the nominative and the accusative of the root *jazyk*, which allow direct lexicalisation of the structure by the root, and therefore do not trigger movement. However, when K3 is merged on top of K2P, the root cannot lexicalise K3P, and movement will be triggered. This will create a constituent $[_{\text{K3P}}$ K3], for which there is no match in the lexicon, however. Although there is a suffix *-a* which can lexicalise K3, the lexical tree of the suffix, (32c), does not contain a constituent $[_{\text{K3P}}$ K3]. At this point, the derivation reaches a dead end, and the backtracking option of the Lexicalisation Algorithm is activated, (25d). The derivation goes back to the point where SG was merged, and instead of taking the first option of the algorithm (direct lexicalisation), it now, on this second pass, takes the second option, and moves the complement, i.e. the NP. The result is a tree that has the constituent $[_{\text{SGP}}$ SG] as its right-hand member, and since this constituent finds a match in (32)(c), it is lexicalised as *-a*, yielding *jazyk-a*. Subsequently, K1 is merged, and again movement of NP will be triggered, followed by lexicalisation of $[_{\text{K1P}}$ K1 $[_{\text{SGP}}$ SG]] by *-a.* The same happens when K2 and K3 are merged, ultimately yielding the genitive form *jazyk-a*, which will have an identical structure to the genitive *chleb-a*, effectively neutralising the distinction between these two different root classes. In this volume, backtracking features in Chapters 2 (Bergsma, Don, Merkuur and Smith), 7 (Demonie), 9 (Caha), 10 (Kasenov), 11 (Caha and Taraldsen Medová), 12 (Taraldsen), and 13 (Balsemin & Pinzin). In the literature, backtracking derivations are discussed, following Starke (2018), in Caha (2019; 2021; 2024), Wiland (2019) or Vanden Wyngaerd et al. (2020).

4.4. Complex specifiers

The final, last-resort option of the algorithm in (25), is the creation of a new workspace, where an entirely new, secondary, derivation is started, which is then subsequently merged with the primary derivation as a prefix or complex specifier, see (25)(e). Complex left branches merged in a new workspace surface as prefixes, or as other material that precedes the root (auxiliaries, complementisers, adverbs, etc). An example of this is the prefix *nej-* that marks the superlative in Czech, as in ***nej**-inteligent-n-ějš-í* 'most intelligent' (see (12)(c) above). Comparing Czech *nej-* to English *-est*, we see two differences. First, *nej-* stacks on top of the combination of the root and the comparative suffix, whereas *-est* attaches directly to the root. Second, *nej-* precedes the root, whereas *-est* follows it. The first of these differences arises because *-est* lexicalises both the CMPR and the SPRL feature, as is obvious from its lexical entry given in (28)(c) above. In contrast, Czech *nej-* lexicalises only SPRLP. If the lexical tree of *nej* had the shape [$_{\text{SPRLP}}$ SPRL], it would be a suffix: since merge is binary, the only way to create a single-branching node like [$_{\text{SPRLP}}$ SPRL] is to generate it as a binary branching one, and move one of the members out. Since movement is always upwards and linearized on the left, *nej*, would end up on the right, as a suffix. The theory thus predicts that suffixes have a single daughter in their lowest branching node (as shown for the Czech Case suffix *-a* in (32c) above), whereas prefixes (and roots) have binary branches in their lowest branching node (compare the root *chleb* 'bread' in (32a) above). This reasoning was proposed in Starke (2018) and is sometimes described as suffixes having a unary bottom, prefixes binary bottoms. Applied to the case of *nej-*, we will assume that it has a lexical tree of the shape [$_{\text{SPRLP}}$ SPRL CMPR], with a binary bottom. When the feature SPRL is merged, it will not be directly lexicalisable by the root, and all the relevant movement options of the algorithm will produce a tree with the unary bottom [$_{\text{SPRLP}}$ SPRL], but there is no match for that in the Czech lexicon. As a last resort, therefore, a new derivation will be spawned in a separate workspace to create a lexicalisable tree with the feature SPRL. In our example, the result is [$_{\text{SPRLP}}$ SPRL CMPR], which matches the lexical entry for *nej*.[4] The main derivation may now merge the tree produced by the secondary workspace and proceed further. More discussion and examples of derivations with complex prefixes can be found in the papers by Wiland (this volume), Demonie (this volume), Gök and Demirok (this volume), Caha (2019) and De Clercq (2020).

As a final point in this section, we want to mention the issue of feature-driven movement, i.e. the syntactic movement of a phrase that has already been lexicalized in an earlier stage of the derivation. Such movements can be incorporated into the lexicalisation algorithm, by allowing the existing derivation to be searched for an already existing lexicalisation of F, either before merge-F or after a failure of merge-F to lexicalise. This is a largely unexplored area in the nanosyntactic literature, but see De Clercq (2019; 2020), De Clercq and Reintges (2022), Caha and Ziková (2023) and Starke (2024) for proposals incorporating feature-driven movement. Incorporation of feature-driven movement into the algorithm points into the direction that Nano is Syntax. We are confident that future work will explore this issue in much greater detail than our summary treatment in the present context.

5. The shape of lexical items

In this section, we focus on two apparent paradoxes for the theory introduced so far, namely *ABA violations and inverse marking systems. Systems of inverse marking have the property that a larger structure is lexicalized by less morphemes than a smaller one. A hypothetical case would be one were the positive degree of an adjective required a morpheme in addition to the root (as in the Czech *intelligent-n-í*), and the comparative of that same adjective would be lexicalized by the root alone (as in the case of Armenian is *partsrahasag* 'taller' discussed above). The lexicalisation algorithm would appear to rule out such a hypothetical situation: if the lexicon contains a root of size CMPRP, then it is never expected to give rise to movement (and consequent suffixation) in the positive degree. We discuss such a pattern from the domain of singular and plural marking in Kipsigis in section 5.1 below. The other paradox is the existence of *ABA patterns in the domain of verbal inflection, which we turn to in section 5.2

We show that both these phenomena can be accounted for thanks to the fact that phrasal lexicalisation allows the lexicalisation of any shape of tree, including trees with "specifiers" (complex left branches). So far, the left branches of lexemes were always terminals, yielding trees of the shape [X [Y [Z]]]. However, if we moved YP across X, the resulting tree would be of the shape [[Y [Z]] X]. The latter is a well-formed syntactic object, and can therefore be stored lexically (Starke 2014a).

Lexically stored trees of this latter type can be referred to as trees with a Complex Left Branch (CLB). The existence of such trees introduces another type of variation in addition to root size. In the example just given, the size of the lexical entries is exactly the same, i.e. the exact same features X, Y, and Z are involved, but the way they are hierarchically organized in the lexical entries is different. The exploration of the possibilities of lexical items with CLBs constitutes the first of the two recent innovations we mentioned in the introduction (the other being the pied-piping-based algorithm, to which we turn in section 6). It turns out that next to root size, root shape can account for certain lexicalisation patterns which would otherwise be unexplained.

5.1. Inverse marking

This section shows that CLB lexical items provide a natural explanation for systems of inverse marking, which raise a paradox stemming from containment (Blix 2022). An example of such a system is the pattern of number marking in Kipsigis (Kalenjin). Table 1.6 summarizes the core of the system of nominal classification that Blix (2022) discusses.

	root	SG	PL	
a.	kipaw	kipaw	kipaw-**tiin**	‘rhino’
b.	peel	peel-**yaan**	peel	‘elephant’
c.	pata	pata-**yaan**	pat-**een**	‘duck’

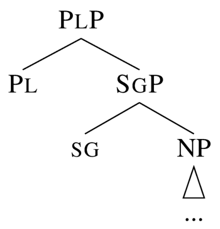
Table 6 Endo-Marakwet (Kalenjin), modified after Blix (2022) via Kouneli (2019, 2021)

The table shows that some nouns -- like *kipaw* ‘rhino’ on row (a) -- show a common pattern where the plural is marked with respect to the singular. Row (b) shows a case where the singular is marked. This type of marking is sometimes referred to as singulative, and the whole pattern can be summarised under the notion of an ‘inverse’: the idea is that each root has a default reference, which is either singular (*kipaw* ‘rhino’) or plural (*peel* ‘elephant’), and the inverse is the marking used when the actual number is the inverse of the default reference. Row (c) shows that there are also nouns where both the singular and plural are marked.

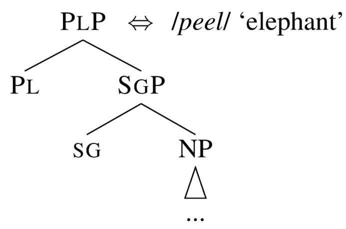
The potential paradox arises because it is impossible to represent both (a) and (b) patterns in terms of morphosyntactic containment because structural containment is asymmetric (i.e., if the structure of the plural properly contains the structure of the singular, it cannot be that at the same time, the singular structure properly contains the plural structure).

To state the problem in terms of the Lexicalisation Algorithm, let us adopt Blix’ functional sequence as in (33)(a) where the plural contains the singular (as in *chair* ~ *chair-s*, or as in row (a) of Table 1.6). Given that *peel* ‘elephant’ on row (b) has no marking in the plural, we would expect that it has an entry as in (33)(b).

(33) a.



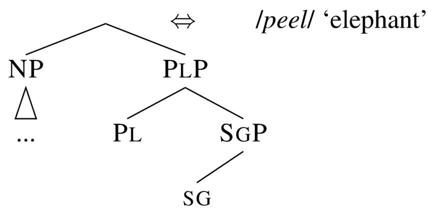
b.



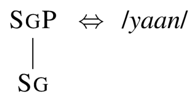
But if that is so, then we would expect that *peel* can also lexicalise the singular number (corresponding to SGP), which is contained inside the lexical tree of *peel*, according to (33)(a).

It turns out that the functional sequence (33)(a) is sufficient to account for the pattern in Table 1.6: If the lexical entry of *peel* contains a complex left branch (NP) arising from an evacuation movement, as in is **Fout! Verwijzingsbron niet gevonden.**(a), the inverse marking is derived. Such a tree arises in the course of the derivation when the NP cyclically moves from below SG (34(a)). Since **Fout! Verwijzingsbron niet gevonden.**(a) is a well-formed structure, and lexical items are links between well-formed syntactic structures and phonology, the theory expects such lexical items to exist. Observe that (33a) and (34a) contain exactly the same features, but arranged in a different shape.

(34) a.



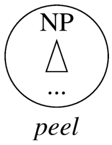
b.

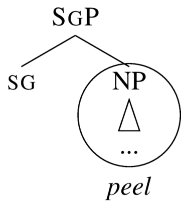


The line connecting SGP with its daughter SG in **Fout! Verwijzingsbron niet gevonden.**(a) is oblique only to visually indicate the extraction site of the NP. It is notationally equivalent to the straight line in **Fout! Verwijzingsbron niet gevonden.**(b), and we shall use both notations in what follows. The lexical item in **Fout! Verwijzingsbron niet gevonden.**(b) represents the singular marker *yaan*. This entry is consistent with the pattern seen on row (c) of Table 1.6, where we see a root that has a marker both for the singular and the plural. Since this root always requires a number marker, we analyse it as lexicalising just an NP, and that is why *yaan* is needed in the singular.

Let us now observe how the markers in **Fout! Verwijzingsbron niet gevonden.** interact with the derivation of the noun *peel* 'elephant.' The first step in the derivation is to assemble the NP, which is lexicalised as *peel*, since NP is contained in the entry **Fout! Verwijzingsbron**

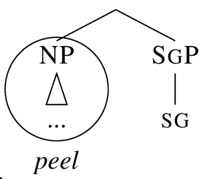
niet gevonden.(a), namely on the left branch. Since matching succeeded, the SG feature is merged, see **Fout! Verwijzingsbron niet gevonden.**(b).

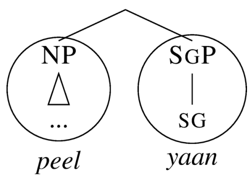
(35) a. 

b. 

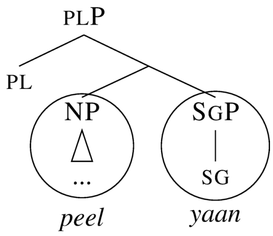
The initial problem with **Fout! Verwijzingsbron niet gevonden.**(b) was that when we postulated the entry for *peel* as in (33)(b), the expectation was that the structure **Fout! Verwijzingsbron niet gevonden.**(b) will be lexicalised by *peel*, yielding the incorrect unmarked singular. However, after we have modified the entry for *peel* to include a complex left branch, the tree in **Fout! Verwijzingsbron niet gevonden.**(b) is actually not contained in the lexical entry **Fout! Verwijzingsbron niet gevonden.**(a): the entry contains no constituent that is identical to **Fout! Verwijzingsbron niet gevonden.**(b). As a consequence, there is no match with **Fout! Verwijzingsbron niet gevonden.**(b) and the Lexicalisation Algorithm resorts to rescue movements.

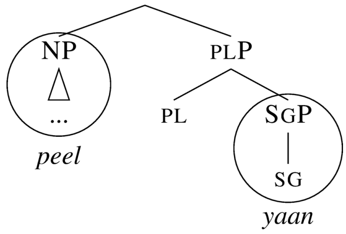
Spec-to-spec movement is inapplicable, complement movement yields **Fout! Verwijzingsbron niet gevonden.**(a). This structure leads to the correct lexicalisation *peel-yaan* ‘elephant-SG’, as in **Fout! Verwijzingsbron niet gevonden.**(b).

(36) a. 

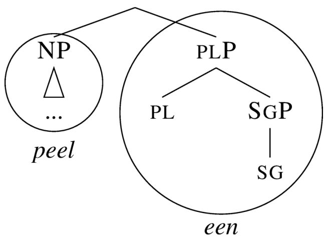
b. 

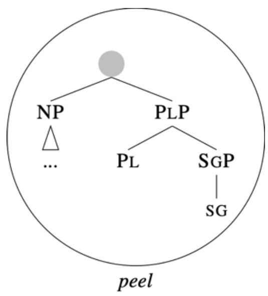
When PL is merged, as in 0(a), there is no match. Spec-to-Spec movement is therefore tried, yielding 0(b).

(37) a. 

b. 

There are two ways to lexicalise 0(b). One option is to use the plural marker *een*, as in **Fout! Verwijzingsbron niet gevonden.**(a). This marker appears in the plural of the noun *pata* 'duck' on row (c) of Table 1.6. Since *pata* 'duck' is just the lexicalisation of an NP, *een* is the realisation of the remaining features.

(38) a. 

b. 

The other option is to use the entire entry of *peel*, as it matches the topmost constituent of the entire structure (i.e. the node marked by a grey circle in **Fout! Verwijzingsbron niet gevonden.**(b)). Since higher lexicalisations override previous ones, this lexicalisation will override the lexicalisation in **Fout! Verwijzingsbron niet gevonden.**(a).

In sum, lexical items with a complex left branch capture 'inverse' patterns of marking, i.e. patterns that seem to give rise to a containment paradox.

5.2. ABA patterns

Another important feature of CLB entries is that they allow for the generation of a special type of ABA pattern, represented as A-Ax-A patterns by Blix (2022). Such a pattern is attested in a dialect of Friulian (spoken in Tualis) studied by Cortiula (2023). In this dialect, the first person is unmarked and syncretic with the third person (*bat* 'I hit/he hits'), while the second adds a suffix to the root (*bat-s* ‘you hit’). When the persons are ordered 1-2-3, this gives rise to an A-Ax-A pattern, as shown in the Friulian column of Table 1.7.

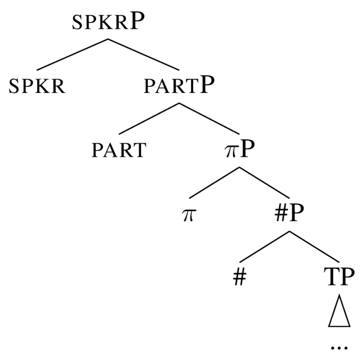
		FRIULIAN *bati* ‘hit’	**BULGARIAN** *četa* ‘read’
SG	1	bat	čet-ox
	2	bat-s	čet-e
	3	bat	čet-e

Table 7 Present tense of *bati* ‘hit’ in Tualis Friulian and aorist of *četa* ‘read’ in Bulgarian

Before we show how CLB trees allow for the generation of this pattern, let us say why Cortiula adopts the ordering 1-2-3, rather than 2-1-3, which would make the Friulian paradigm trivially compatible with *ABA. The reason is that in other languages, of which Bulgarian in the second column of Table 1.7 is an instance, we find a syncretism between second and third person, which are incompatible with the 2-1-3 order. The syncretism between second and third person is in fact more common than the one between first and third person (even though the latter is far from being rare either; see Baerman et al. 2005; Baerman and Brown 2013; Ackema and Neeleman 2013). Cortiula (2023) therefore adopts the person ordering 1-2-3.

The specific decomposition of the agreement features she works with is given in (39) (see Béjar 2003, Starke 2020, Blix 2021a, Starke and Cortiula 2021).

(39)



Let us start unpacking this structure by pointing out that it is essentially an extended version of the proposal by Pollock (1988), who proposed that above the tense projection, there is a dedicated position for agreement features. In (39), each agreement feature is given an independent projection, in accordance with the general tenets of the Nanosyntax approach. Let us now introduce the ingredients.

Immediately above TP, we see the number projection #. Plural would require the addition of PL, as discussed in the previous section, but we ignore this here for simplicity. In the absence of PL, number is interpreted as the default number, which is singular. Above number are person features. If only the feature π was present, we would be looking at a default person, i.e., the third person. The second person has the additional feature PART(icipant), and first person is semantically most complex, also including the feature SPKR. With respect to this decomposition, the paradigm found in Friulian represents a *ABA paradigm.

The fact that the form *bat* can lexicalise the first-person structure suggests that it is able to lexicalise all the features present in (39). Given the matching condition (18), this allows it to lexicalise the third-person, which is structurally contained in the first person. What is unexpected is that the root *bat* needs the suffix *-s* in the second person: by standard logic of constituent matching, the root should be able to lexicalise PARTP on its own.

In the form of a lexicalisation table, the pattern looks as in Table 1.8.

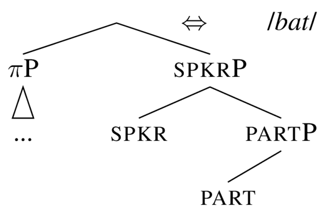
	IndP	T	#	π	PART	SPKR
1SG	bat					
2SG	bat				s	
3SG	bat					

Table 8. Lexicalisation table of the present tense singular in TF

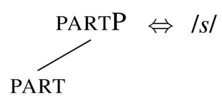
This looks like 'selective shrinkability': a large lexical item A cannot shrink one notch (requiring help of another lexical item B instead), but can shrink two notches. We indicate this nonshrinkability by means of the asterisk in Table 1.8. Applied to the case at hand, the root *bat* can lexicalise a constituent as large as SPKRP (1SG). It can also lexicalise the subconstituent πP of 3SG, but it cannot lexicalise the subconstituent PARTP (2SG), since it needs the suffix *-s* to lexicalise the feature PART. The inability of the root to lexicalise PARTP is what the asterisk in Table 1.8 indicates.

Lexicalising a structure which includes an evacuation movement however yields exactly that distribution. Cortiula proposes the lexical entry in **Fout! Verwijzingsbron niet gevonden.**(a) for the root *bat,* and the one in in **Fout! Verwijzingsbron niet gevonden.**(b) for the 2nd person suffix *-s*.

(40) a.



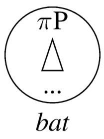
b.

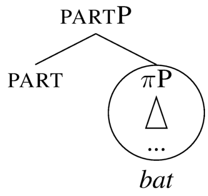


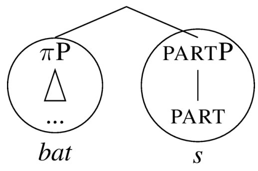
This lexical entry with the CLB can lexicalise the most complex structure (SPKRP), but it does not contain a subconstituent that is identical to the structure of a 2SG: the PARTP in **Fout! Verwijzingsbron niet gevonden.**(a) only dominates PART, and not the rest of the features that would be needed for the root to lexicalise the full 2SG form of the root. On the other hand, the entry does contain a subconstituent as a CLB, namely πP, which corresponds with the least complex form (3SG). In the lexicalisation table (Table 1.8), the feature that is immediately to the left of the asterisk corresponds with the size of the CLB.

Let us now go through the derivation stepwise. The derivation first takes several steps to construct the πP, corresponding to the third person. Since πP is contained in the entry of *bat* in **Fout! Verwijzingsbron niet gevonden.**(a) as a CLB, the third person is lexicalised as *bat*, as in **Fout! Verwijzingsbron niet gevonden.**(a). When the PART feature is merged, as in **Fout!**

Verwijzingsbron niet gevonden.(b), there is no match. Therefore, lexicalisation-driven movement is required. Spec-to-Spec movement does not apply as there is no such phrase to be moved, so that complement movement applies, yielding **Fout! Verwijzingsbron niet gevonden.**(c), which is lexicalised as *bat-s*.

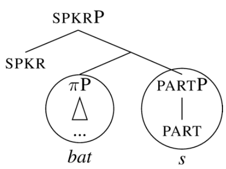
(41) a. 

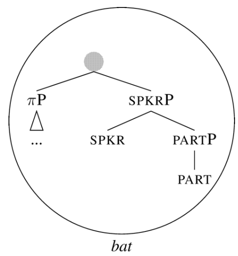
b. 

c. 

Note that **Fout! Verwijzingsbron niet gevonden.**(c) cannot be lexicalised as *bat*, since there is no constituent in the lexical entry of *bat* in **Fout! Verwijzingsbron niet gevonden.**(a) that is identical to **Fout! Verwijzingsbron niet gevonden.**(c).

To derive the first person, the SPKR feature is merged, as in **Fout! Verwijzingsbron niet gevonden.**(a). Since there is no match, spec-to-spec movement is triggered, yielding **Fout! Verwijzingsbron niet gevonden.**(b). This constituent is identical to the lexical tree of *bat*, and the first person is therefore lexicalised as *bat*.

(42) a. 

b. 

In sum, lexical entries with CLBs have been understudied until Blix's ground-breaking work (Blix 2021b), but once the spotlight is shone on them, they derive important morpho-syntactic patterns. Their empirical relevance has been demonstrated in the domain of Czech declension classes by Janků (2022), that of Friulian verb classes (Cortiula 2023), and the morphology of Czech adjectives (Caha, De Clercq and Vanden Wyngaerd to appear; Vanden Wyngaerd, De

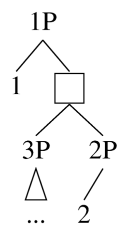
Clercq and Caha to appear). In this volume, CLB entries are explored in the contributions by Bergsma, Don, Merkuur and Smith, by Caha, by Caha and Taraldsen Medová, by Kasenov, and by Vyshnevska.

6. A new logic for the lexicalisation algorithm

Starke (2022) suggests a novel way of thinking about rescue movements (evacuation movements), with the side-effect that a new type of evacuation becomes available: subextractions. The core of the idea is that rescue movements involve gradual pied-piping: Evacuation always targets the same constituent, but that constituent can pied-pipe gradually larger structures in the quest for a successful lexicalisation.

Let's unpack that by looking at a typical derivation such as **Fout! Verwijzingsbron niet gevonden.**.

(43)



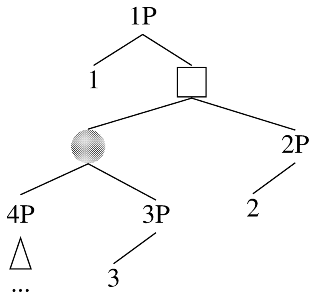
Here 3P has evacuated out of 2P, and the square node lexicalised successfully. Now feature 1 has been merged and the constituent cannot lexicalise as it is. Starke's reformulation of the lexicalisation algorithm always evacuates 3P in this configuration. That leads to a spec-to-spec evacuation. If this evacuation fails to yield a lexicalisation, 3P retries by pied-piping its mother node, i.e. the square constituent. This formulation derives both the existence of complement movement, without the need to stipulate that there is such an option and the fact that it comes after spec-to-spec evacuation, without having to explicitly list that order.

Why 3P though? Two answers have been given, one in Starke's original 2022 seminar and one in Caha's follow up 2023 seminar. In both formulations, only labelled nodes can be targeted by movement.

Starke's proposal is a generalisation of the insight in Cinque (2005) that only lexical categories move in these types of movements (in Cinque's case, only nouns move), together with the syntactic locality principle of moving the closest candidate. In the syntactic tradition (and in Cinque's system), "lexical categories" are the bottom of the clause, the bottom of *fseq*. The condition of moving only nouns, thus generalises to 'move a node containing the bottom of *fseq*'. If we combine that with 'move the candidate closest to the root of the tree', we obtain that in (43), the node 3P is the closest node (to the root of the tree) containing the bottom of *fseq*. Starke's proposal is thus that evacuation always targets the closest labelled node containing the bottom of the *fseq*, and if that fails to lexicalise, that same node pied-pipes its containing node.

In Caha's formulation, the same effect is obtained by restricting the target to the closest labelled non-remnant constituent. In (42) for instance, 3P is the only non-remnant node since 2P is a remnant in the sense that material has been extracted out of it (see Caha and Taraldsen Medová, this volume; Caha, this volume, for more details).

As a side-effect, this new way of thinking about evacuation movements allows for a new type of rescue movement: subextraction. Consider the tree in **Fout! Verwijzingsbron niet gevonden.**, a stage of the derivation where 4P has evacuated out of 3P, the resulting circled node has evacuated out of 2P, and the feature 1 is then merged.

(44) 

Since only labelled nodes may be evacuation targets, 2P, 3P and 4P could in principle be evacuated, but the round or square nodes could not. In this situation, both Starke's and Caha's formulation have the effect that 4P ends up being the only candidate for evacuation. This then yields a new evacuation type: subextraction from within a "Spec". If the evacuation of 4P fails to create a lexicalisable structure, the mother node of 4P, marked by the gray circle, would be pied-piped, yielding a traditional spec-to-spec evacuation. If that in turn fails to create a lexicalisable structure, another degree of pied-piping is attempted, evacuating the entire square node – yielding a traditional complement movement. The new formulation thus deduces both the types of evacuation movements, and their relative ordering, while predicting new types of evacuation movement: sub-extraction, sub-sub-extraction, etc.[5]

By doing so, this new version of the algorithm preserves previous analyses, as it derives the same movements (spec-movement, the gray circle, and complement movement, the square), in the same order (always moving the spec before moving the complement). In addition to being more principled, preserving previous accounts, and predicting the existence of subextractions, this algorithm gives us new analytical options. Let's show this with a small case study of the Friulian conjugation, using Caha's formulation of the algorithm for concreteness:

(45) (a) Merge F and lexicalise

(b) If fail, evacuate the closest labelled non-remnant constituent and lexicalise

(c) If fail, evacuate the immediately dominating constituent and lexicalise (recursive)

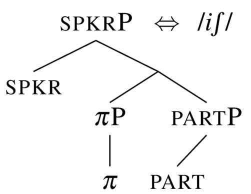
The Friulian conjugation of *bati* ‘to hit’ and *finî* ‘to finish’ are shown in Table 1.9.[6]

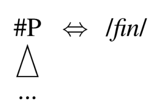
		FRIULIAN	
		fini ‘finish’	*bati* ‘hit’
SG	1	fin-iʃ	bat
	2	fin-iʃ-s	bat-s
	3	fin-iʃ	bat

Table 9 Present tense singular of *bati* ‘hit’ and *finî* ‘finish in Tualis Friulian

The table shows that the two verb classes have the same overall pattern: there is no phi feature suffix in the first and third persons, while *-s* appears in the second person. The difference is that this happens after a single root in *bati* ‘hit’ whereas it happens after a multi-morphemic stem in *fini* ‘finish’. This difference makes the technical derivation for *bati* ‘hit’ inapplicable for *fini* ‘finish’. The new algorithm however allows a new derivational path, yielding R-a/R-a-x/R-a patterns (where R represents the root, and -a represents any number of suffixes).

The lexical entry for *iʃ* contains a CLB, as shown in **Fout! Verwijzingsbron niet gevonden.**(a), just like the lexical entry for *bat* in **Fout! Verwijzingsbron niet gevonden.**(a) did.

(46) a. 

b. 

Notice an important difference: the entry for *iʃ* does contain a constituent corresponding to the second person (π and participant), whereas the crux of the derivation of *bati* was that it does not contain any such constituent. Despite this, **Fout! Verwijzingsbron niet gevonden.**(a) will fail to lexicalise the second person, due to how the derivation is timed.

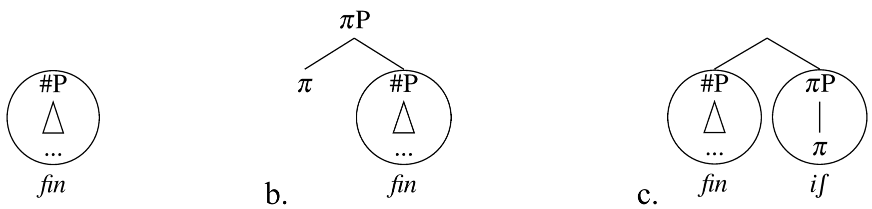
Simplifying Cortiula's account, we posit the entry for the root *fin-* 'finish' as in (45)(b). This entry can only lexicalise features up to #P. Unlike *bat* 'hit,' *fin* cannot lexicalise πP, which is why *-iʃ* has to appear. The lexicalisation table corresponding to this analysis is provided in Table 1.10. The table shows (with an asterisk on the line of 1SG) the special shape of the lexical entry for *-iʃ*, which can shrink to the size of its left branch (πP), but not to the size of PARTP.

	TP	#	π	PART	SPKR
1SG	fin		-iʃ	*	
2SG	fin		-iʃ	s	
3SG	fin		-iʃ		

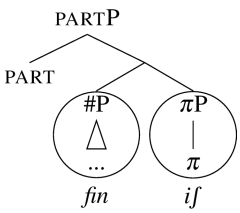
Table 10 Lexicalisation table of TF singular present tense of *iʃ*-verbs

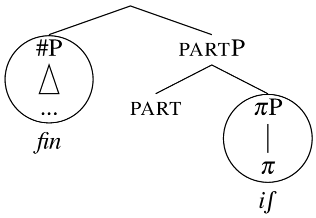
The derivation starts by assembling #P, see (47)(a). When πP is merged, as in (47)(b), *fin* is no longer a match. Following the Lexicalisation Algorithm, we must subextract the node which is closest to the root node, and which is at the same time labelled and non-remnant. In (47)(b), this is the #P node. After #P is evacuated from πP, we get the correct third person *fin-iʃ*, see (47)(c).

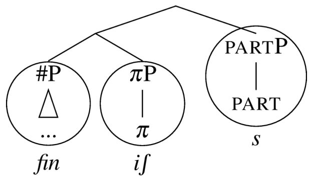
(47) a. b. c.



The derivation continues by merging PART, see (48)(a). Since there is no match, we must evacuate the closest non-remnant labelled node. This cannot be the complement of PART (since it is unlabelled), and it cannot be πP (since it is remnant), so it must be #P. Evacuating #P gives us (48)(b), but there is no match for PARTP, since the entry for *-iʃ* does not contain this constituent.

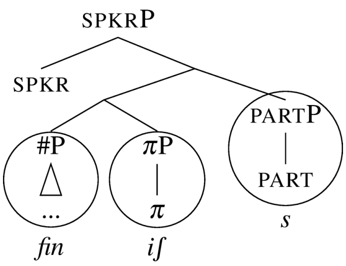
(48) a. 

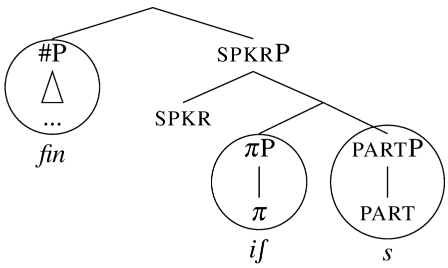
b. 

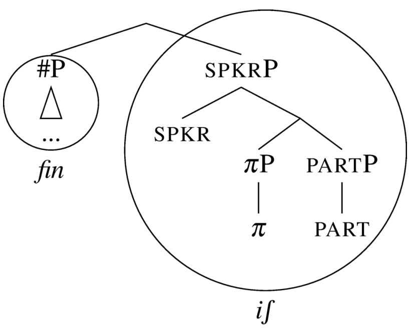
c. 

As a result, we revert to (48)(a), and we pied-pipe the node immediately dominating #P, which is the complement of PART. Evacuation of this node yields (48)(c), which can be successfully lexicalised, yielding the correct second person form.

Notice that so far, the pied-piping algorithm gives the exact same steps as the standard algorithm: we first moved the spec, and then the complement. The difference between the algorithms surfaces when we introduce the speaker feature, as in (49)(a).

(49) a. 

b. 

c. 

Since there is no match for (49)(a), we need to move the closest non-remnant labelled constituent. Looking at (49)(a), this is not the spec of the complement, which has no label. Hence, at this point, the difference between the old and the new algorithm emerges, because the subextracting algorithm leads to the evacuation of #P. This is shown in (49)(b). We see that this movement strands the πP in between SPKR and PART, which is precisely the shape of the entry for *-iʃ*, which has a remnant πP stranded between these two features. As a result, the whole SPKRP can be lexicalised as *-iʃ* (49)(c). This derives the ABA-pattern displayed by the *-iʃ*-suffix in Tualis Friulian.

For further explorations of the revised algorithm, we refer the reader to Kasenov (2023), as well as Caha and Taraldsen Medová (this volume) and Caha (this volume).

7. Comparison with other approaches

It is beyond the scope of this chapter to provide a full and detailed comparison of the nanosyntactic approach with other approaches to syntax and morphology. We wish nevertheless to devote a few comments to this issue, in particular with respect to other approaches which we take to be particularly relevant to the current discussion. Keeping to the innovations in the Lexicalisation Algorithm, and the one involving their shape in particular, we can see an important difference with approaches that assume less restrictive formulations of the principles that determine when a lexical entry can lexicalise a given syntactic structure. We saw earlier (section 5) that the Matching Condition imposes not just a match in the number and type of features that lexical and syntactic representations must contain, but also a structural match in terms of constituent identity. This implies that a lexical tree of the shape [[Y [Z]] X] is not a match for the syntactic structure [X [Y [Z]]], even though the lexical tree and the syntactic tree have exactly the same set of features (namely, X, Y, and Z). A less restrictive alternative ignores the shape of the lexical entry, and just considers the question whether the set of features stand in a subset-superset relationship, taking this to be a sufficient condition for lexicalisation. Such an alternative is proposed in Vanden Wyngaerd (2018), under the name of the Revised Superset Principle, which is used in the contribution by Baunaz and Lander (this volume). It is to be noted, however, that the types of phenomena discussed in section 5, like the Kipsigis inverse marking and the Friulian ABA patterns, cannot be accounted for under this alternative, less restrictive, approach in terms of the Revised Superset Principle. Crucially, a lexical entry that would contain the features X, Y, and Z could – under such an approach - always lexicalise a syntactic structure containing those features (or a subset of them), regardless of their structural arrangement.

Another approach uses spans rather than constituents (e.g. Ramchand 2008; Svenonius 2012, 2016; Merchant 2015). Spans are "a complement sequence of heads, normally in a single extended projection" (Svenonius 2012:1). Taking again the structure [X [Y [Z]]], the complement sequence of heads X-Y-Z (and any contiguous subpart thereof) is a span. In contrast, in the structure [[Y [Z]] X], with the (moved) constituent [Y [Z]], there is a span Y-Z and a (trivial) span X, but no span X-Y-Z, since the (moved) constituent is not the complement of X, nor the other way round. The spanning approach takes morphemes to be lexicalisations of spans: "[a] single morphological exponent (morpheme, for short) cannot spell out two heads (cannot "span" two heads) unless those heads are in a complement relation with each other" (Svenonius 2012: 2). This means that the results of section 5 cannot be derived in a spanning approach. To see this, consider the Kipsigis pattern of inverse number marking, where a noun like *peel* 'elephant' needs a suffix to mark singular number, but none to mark the plural. In our approach, this was done by assigning to such nouns a lexical entry with a CLB (see (33a) above), i.e. [[PL [SG]] N]. This allowed the root to lexicalise the features N, SG, and PL, while at the same time preventing it from realising the features N and SG in the singular, requiring a suffix to lexicalise SG. In the spanning approach, lexicalisation of the plural by the root would require a span PL-SG-N (corresponding to a syntactic structure [PL [SG [N]]]), but a root with such a lexical entry would always be able to lexicalise the singular, and the need for a suffix in the singular would be unexplained.

Distributed Morphology (DM) is another framework using late-insertion (on top of early insertion of language-specific "bundles"). In contrast to Nanosyntax, it does not (usually) use phrasal lexicalisation, and instead all morphemes are inserted under terminal nodes. This means that an approach to allomorphy where root size conditions allomorphy, is unavailable since variable (syntactic) root size requires phrasal lexicalisation. Instead, DM takes extensive recourse to zero morphemes, and rules of contextual allomorphy to ensure their correct distribution. The combination of terminal spellout, zero morphemes, and rules of contextual allomorphy can also deliver the patterns of inverse marking, like the one we discussed for Kipsigis. However some of these patterns of distribution cannot be derived in DM without the postulation of accidentally homophonous morphemes, as argued in Caha (2024), Caha et al. (to appear) and Vanden Wyngaerd et al. (to appear).

ABA-patterns of the Friulian type are difficult to account for if the same decomposition of the agreement marker as in (39) above is adopted, with a set of features that stand in a containment relationship. A similar containment is crucial in Bobaljik’s (2012) account for the absence of ABA-patterns in root suppletion in the triplet positive-comparative-superlative. Adopting such containment in the person-number sequence will make it consequently hard to allow the ABA-like distribution of the agreement marker. The fact that such patterns are widespread in verbal agreement marking (as noted above) may therefore lead the DM-theorist to abandon the containment hierarchy of (39), and in this manner develop an account for the occurrence of ABA-patterns (see e.g. Harbour 2008; 2016, and references cited there). Discussing the various avenues where this may lead is clearly beyond the scope of the present chapter. However, Taraldsen’s chapter in the current volume compares Nanosyntax to

Distributed Morphology in his analysis of suppletive verbs in Romance. We also refer the reader to Caha (2018, to appear), and Barbiers, van der Wal and Vanden Wyngaerd (to appear), for more general discussion of the differences between DM and Nanosyntax.

8. Conclusion

This chapter has outlined the basic assumptions of Nanosyntax, focusing in particular on the Lexicalisation Algorithm. In addition to the basic lexicalisation-driven movements, the backtracking and complex specifier options of the Algorithm were shown to be a means of accounting for the phenomenon of paradigmatic levelling and prefixes, respectively. In addition, this chapter zoomed in on two recent evolutions within the framework. The first concerns the possibilities offered by lexical items with Complex Left Branches (CLBs), which considerably extend the empirical coverage of the nanosyntactic framework. The second evolution is a new logic for the Lexicalisation Algorithm proposed in Starke (2022), which simplifies it and as a side-effect adds subextractions as a type of rescue movement.

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Notes

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[1] In Cinque (2005), the NP moves to the Spec of intermediate Agr projections. We ignore this here for simplicity.

[2] The chapter by Balsemin and Pinzin discusses an interesting case where the winner is determined by phonology, rather than by the Elsewhere Principle.

[3] See Caha et al. (2019); Vanden Wyngaerd et al. (2021) for a discussion of how a particular root is selected among those that match. One idea is that roots do not compete in terms of the Elsewhere Condition, and one inserts the root that "one wants to talk about". This can be implemented by proposing that the conceptual representation restricts the candidate set of competing items, so that the root *tall* is the only root in the competition set (all other roots are irrelevant, since they are linked to the wrong concept).

[4] There is some variation in the literature concerning which other feature combines with the newly required feature: some sources combine the last successfully lexicalized feature with the newly required feature, while other sources combine the newly required feature with the next in line. The latter option would require a decomposition of SPRL into two distinct features (as

proposed in De Clercq and Vanden Wyngaerd 2017). This issue is currently investigated and there is no final consensus on this topic.

[5] Subextractions are a well-known syntactic type of movement, their use in nanosyntax continues the research programme of unifying syntax, morphology and beyond, by reusing and adapting standard operations and filters on structures. Within Nanosyntax, Wiland (2019) already proposed a subextraction step in the structure-building algorithm, ordering it after complement movement however. See also Pantcheva (2011) for a rather different derivational system, within a larger umbrella of Nanosyntactic investigations.

[6] The table ignores certain phonological processes which convert the underlying form of the 2SG into the surface form *finis.* See Cortiula (2023) for discussion.